

|  <b>YTU – Faculty of Chemical and Metallurgical Engineering, Questions and Answers Sheet</b>                                   |                                | NOTE CHART |      |      |              |      |               |           |                  |            |       |       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|------------|------|------|--------------|------|---------------|-----------|------------------|------------|-------|-------|
|                                                                                                                                                                                                                |                                | 1. q       | 2. q | 3. q | 4. q         | 5. q | 6. q          | 7. q      | 8. q             | 9. q       | 10. q | Total |
| Student Name and Surname                                                                                                                                                                                       |                                |            |      |      |              |      |               | X         | X                | X          | X     |       |
| Students Number                                                                                                                                                                                                |                                |            |      |      |              |      |               |           |                  |            |       |       |
| Section                                                                                                                                                                                                        |                                |            |      |      |              |      |               | Exam Date |                  | 18/01/2016 |       |       |
| Course Name                                                                                                                                                                                                    | LINEAR ALGEBRA                 |            |      |      | Group Number |      | Exam Duration | 80 mins.  | Examination Room |            |       |       |
| Course Instructor Name and Surname                                                                                                                                                                             | Assoc. Prof. Fatma AYDIN AKGUN |            |      |      |              |      | Signature     |           |                  |            |       |       |
| Student Disciplinary Regulations "and to make or attempt to make copies of exams to" the actual perpetrators are suspended from one or two semesters. (YÖK; 2547 Student Disciplinary Regulations, 9. Article) |                                |            |      |      |              |      |               |           |                  |            |       |       |

### Linear Algebra Make-up (Final) Exam

Q1)  $A = \begin{bmatrix} 1 & -2 & 1 \\ 1 & 3 & 5 \end{bmatrix}$  and  $X = \begin{bmatrix} x & a \\ y & b \\ z & c \end{bmatrix}$  are given. If  $AX = I$ , then find the matrix  $X$ . ( $I$  is an identity matrix)

Q2)  $\left. \begin{array}{l} x + 2y - 3z = 4 \\ 3x - y + 5z = 2 \\ 4x + y + (k^2 - 14)z = k + 2 \end{array} \right\}$  Equation system is given.

- If the system has **no solution**, then find the appropriate value of  $k$
- If the system has **an unique solution**, then find the appropriate value of  $k$
- If the system has **infinitely many solutions**, then find the appropriate value of  $k$

Q3)  $\vec{a} = \begin{bmatrix} 2m \\ -1 \\ -1 \end{bmatrix}$ ,  $\vec{b} = \begin{bmatrix} -1 \\ 2m \\ -1 \end{bmatrix}$  and  $\vec{c} = \begin{bmatrix} -1 \\ -1 \\ 2m \end{bmatrix}$  are given.

What must be the value of  $m$  if  $\vec{a}, \vec{b}$  and  $\vec{c}$  are linearly dependent? Obtain the relation between  $\vec{a}, \vec{b}$  and  $\vec{c}$ .

Q4)  $A = \begin{bmatrix} 1 & -4 & 2 \\ -4 & 1 & -2 \\ 2 & -2 & -2 \end{bmatrix}$  is given. Find the eigenvalues and eigenvectors of the matrix  $A$ .

Q5) In  $V = P_2$  Euclidean space, prove that  $\langle p(t), q(t) \rangle = \int_0^1 p(t)q(t)dt$  defines inner product for  $p(t), q(t) \in P_2$

Q6) In  $R^3$  space, vectors  $\vec{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ ,  $\vec{v}_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$ ,  $\vec{v}_3 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$  are given.

- Show that  $S = \{v_1, v_2, v_3\}$  is a base for  $R^3$
- Find the orthonormal base  $E = \{g_1, g_2, g_3\}$  for  $R^3$

Only **four** questions must be answered and you can choose any four questions out of six..

Good Luck!

$$(1) \quad A = \begin{bmatrix} 1 & -2 & 1 \\ 1 & 3 & 5 \end{bmatrix} \quad X = \begin{bmatrix} x & a \\ y & b \\ z & c \end{bmatrix}$$

$$AX = \begin{bmatrix} x-2y+z & a-2b+c \\ x+3y+5z & a+3b+5c \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad (5)$$

$$\begin{aligned} x-2y+z &= 1 \\ x+3y+5z &= 0 \end{aligned} \quad \left[ \begin{array}{ccc|c} 1 & -2 & 1 & 1 \\ 1 & 3 & 5 & 0 \end{array} \right] \sim \left[ \begin{array}{ccc|c} 0 & -5 & -4 & 1 \\ 1 & 3 & 5 & 0 \end{array} \right]$$

$$x_3 = t \quad x_2 = \frac{1+4t}{-5}$$

$$\begin{aligned} x_1 + 3x_2 + 5x_3 &= 0 \\ x_1 + \frac{3+12t}{-5} + 5t &= 0 \end{aligned}$$

$$x_1 = -\frac{13t+3}{5}$$

$$(10)$$

$$a-2b+c=0$$

$$a=2b-c$$

$$a+3b+5c=1$$

$$5b+4c=1$$

$$c = \frac{1-5b}{4}$$

$$b = \frac{1-4c}{5}$$

$$a = \frac{2-8c}{5} - c$$

$$a = \frac{2-13c}{5}$$

$$(10)$$

$$X = \begin{bmatrix} \frac{-13t+3}{5} & \frac{2-13c}{5} \\ \frac{1+4t}{-5} & \frac{1-4c}{5} \\ t & c \end{bmatrix}$$

$$X = \begin{bmatrix} +3/5 & 2/5 \\ -1/5 & 1/5 \\ 0 & 0 \end{bmatrix}$$

$$b=1 \quad c=1$$

$$2) \begin{bmatrix} 1 & 2 & -3 & 4 \\ 3 & -1 & 5 & 2 \\ 4 & 1 & k-14 & k+2 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -3 & 4 \\ 0 & -7 & 14 & -10 \\ 0 & -7 & k-2 & k-14 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 2 & -3 & 4 \\ 0 & -7 & 14 & -10 \\ 0 & 0 & k-16 & k-4 \end{bmatrix} \quad \text{a) } \begin{matrix} k^2-16=0 & k-4 \neq 0 \\ k \neq 4 & k \neq 4 \end{matrix} \quad (5)$$

$k = -4$  dır. ( $k = -4$  ise çözüm yoktur)

b)  $k \neq -4$  ise tek çözüm vardır. (5)

c)  $k = 4$  ise sonsuz çözüm vardır. (5)

$$\boxed{c_1 \vec{a} + c_2 \vec{b} + c_3 \vec{c} = 0} \quad (1)$$

3) 
$$\begin{vmatrix} 2m & -1 & -1 \\ -1 & 2m & -1 \\ -1 & -1 & 2m \end{vmatrix} = 0 \quad \text{abw. d. d.} \quad (5)$$

$$\begin{vmatrix} 2m-2 & 2m-2 & 2m-2 \\ -1 & 2m & -1 \\ -1 & -1 & 2m \end{vmatrix} = \begin{vmatrix} 2m-2 & 0 & 0 \\ -1 & 2m+1 & 0 \\ -1 & 0 & 2m+1 \end{vmatrix}$$

$$= (2m-2)(2m+1)^2 = 0$$

$m_1 = 1, \quad m_{2,3} = -\frac{1}{2} \quad \text{abw. d. d.} \quad (10)$

$m_1 = 1$  ist  $c_1 \begin{bmatrix} 2 \\ -1 \\ -1 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix} + c_3 \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$$\begin{bmatrix} 2 & -1 & -1 & 0 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 2 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & -2 & 0 \\ 0 & 3 & -3 & 0 \\ 0 & -3 & 3 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\vec{a} + \vec{b} + \vec{c} = 0} \quad (5)$$

$$\begin{cases} c_1 - c_3 = 0 \\ c_2 - c_3 = 0 \end{cases} \Rightarrow \begin{cases} c_3 = s \\ c_1 = s \\ c_2 = s \end{cases} \quad (s = 1 \text{ ist m.})$$

oder  $c_1 = 1, c_2 = 1, c_3 = 1$

$m_{2,3} = -\frac{1}{2}$  ist  $\begin{bmatrix} -1 & -1 & -1 & 0 \\ -1 & -1 & -1 & 0 \\ -1 & -1 & -1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$$\boxed{-2\vec{a} + \vec{b} + \vec{c} = 0} \quad (5)$$

$$c_1 + c_2 + c_3 = 0 \Rightarrow \begin{cases} c_2 = 1 \text{ k. f. f.} \\ c_3 = 1 \text{ k. f. f.} \\ c_1 = -2 \text{ oder} \end{cases}$$

$$4) A = \begin{bmatrix} 1 & -4 & 2 \\ -4 & 1 & -2 \\ 2 & -2 & -2 \end{bmatrix} \quad (A - \lambda I)X = 0$$

$$\left. \begin{aligned} (1-\lambda)x_1 - 4x_2 + 2x_3 &= 0 \\ -4x_1 + (1-\lambda)x_2 - 2x_3 &= 0 \\ 2x_1 - 2x_2 + (-2-\lambda)x_3 &= 0 \end{aligned} \right\}$$

$$|A - \lambda I| = 0 \quad (5)$$

$$\begin{vmatrix} 1-\lambda & -4 & 2 \\ -4 & 1-\lambda & -2 \\ 2 & -2 & -2-\lambda \end{vmatrix} = \begin{vmatrix} -3-\lambda & -4 & 2 \\ -3-\lambda & 1-\lambda & -2 \\ 0 & -2 & -2-\lambda \end{vmatrix} = \begin{vmatrix} -3-\lambda & -4 & 2 \\ 0 & 5-\lambda & -4 \\ 0 & -2 & -2-\lambda \end{vmatrix}$$

$$= (-3-\lambda)[\lambda^2 - 3\lambda - 18] = 0$$

$$\lambda_1 = 6, \lambda_2 = -3, \lambda_3 = -3 \text{ bulunur. } (10)$$

$$\lambda_1 = 6 \text{ için } \left. \begin{aligned} -5x_1 - 4x_2 + 2x_3 &= 0 \\ -4x_1 - 5x_2 - 2x_3 &= 0 \\ 2x_1 - 2x_2 - 8x_3 &= 0 \end{aligned} \right\} \begin{bmatrix} -5 & -4 & 2 & 0 \\ -4 & -5 & -2 & 0 \\ 2 & -2 & -8 & 0 \end{bmatrix} \sim$$

$$\sim \begin{bmatrix} 1 & -1 & -4 & 0 \\ 0 & -9 & -18 & 0 \\ 0 & -9 & 18 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{aligned} x_1 - 2x_3 &= 0 & x_3 = 5 \\ x_2 + 2x_3 &= 0 & x_2 = -25 \\ & & x_1 = 5 \end{aligned}$$

$$\lambda_{2,3} = -3 \text{ için } \left. \begin{aligned} 4x_1 - 4x_2 + 2x_3 &= 0 \\ -4x_1 + 4x_2 - 2x_3 &= 0 \\ 2x_1 - 2x_2 + x_3 &= 0 \end{aligned} \right\} \begin{bmatrix} 4 & -4 & 2 & 0 \\ -4 & 4 & -2 & 0 \\ 2 & -2 & 1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 1/2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$X_{2,3} = s \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix} + t \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad \text{bulunur} \quad (5)$$

$$\begin{aligned} x_1 - x_2 + \frac{1}{2}x_3 &= 0 \\ x_3 &= 25 \\ x_2 &= t \\ x_1 &= t - 5 \text{ olur.} \end{aligned}$$

$$(5) \quad \langle p(t), q(t) \rangle = \int_0^1 p(t) q(t) dt$$

$$(1) \quad \langle p(t), p(t) \rangle \geq 0 \quad \langle p(t), p(t) \rangle = 0 \Leftrightarrow p(t) = 0$$

$$\int_0^1 |p(t)|^2 dt \geq 0. \quad \langle p(t), p(t) \rangle = 0 \Leftrightarrow p(t) = 0.$$

$$\int_0^1 \underbrace{|p(t)|^2} dt = c \Big|_0^1$$

(10)

$$(2) \quad \langle p(t), q(t) \rangle = \langle q(t), p(t) \rangle$$

$$\int_0^1 p(t) q(t) dt = \int_0^1 q(t) p(t) dt$$

(5)

$$(3) \quad k \langle p(t), q(t) \rangle = \langle kp(t), q(t) \rangle \quad (k \text{ sct})$$

(5)

$$k \int_0^1 p(t) q(t) dt = \int_0^1 \underbrace{k p(t)} q(t) dt = \langle kp(t), q(t) \rangle$$

$$(4) \quad \langle p(t) + q(t), w(t) \rangle = \langle p(t), w(t) \rangle + \langle q(t), w(t) \rangle$$

(5)

$$\int_0^1 (p(t) + q(t)) w(t) dt = \int_0^1 p(t) w(t) dt + \int_0^1 q(t) w(t) dt$$

$$= \int_0^1 \underbrace{(p(t) w(t) + q(t) w(t))} dt$$



6) a) 1)  $\alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 = \vec{0}$

2)  $\alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 = \vec{0} \Leftrightarrow \alpha_1 = \alpha_2 = \alpha_3 = 0$

Her iki durumda da çözümü olduğu, 2. homojen sistem  
tek çözümü olduğu

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$\begin{vmatrix} 1 & -1 & 1 \\ 1 & 1 & 2 \\ 1 & 0 & 1 \end{vmatrix} = -1 \neq 0$  tek çözüm. Dolayısıyla bizzat.

b)  $u_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$   $u_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$   $u_3 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \rightarrow$  ortogonal bir

$u_1 = v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$   $v_2 = u_2 - \frac{(u_2, v_1)}{(v_1, v_1)} v_1 = u_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$

$v_3 = u_3 - \frac{(u_3, v_2)}{(v_2, v_2)} v_2 - \frac{(u_3, v_1)}{(v_1, v_1)} v_1$

$= \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} - \frac{4}{3} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

$= \begin{bmatrix} 1 + 1/2 - 4/3 \\ 2 - 1/2 - 4/3 \\ 1 - 0 - 4/3 \end{bmatrix} = \begin{bmatrix} 3/2 - 4/3 \\ 3/2 - 4/3 \\ -2/6 \end{bmatrix} = \begin{bmatrix} 1/6 \\ 1/6 \\ -2/6 \end{bmatrix}$

$w_1 = \begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$

$w_2 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$

$w_3 = \begin{bmatrix} 1/\sqrt{6} \\ 1/\sqrt{6} \\ -2/\sqrt{6} \end{bmatrix}$

